Lab 3: Component Modelling & Architectural Pattern Selection

NAME: TRISHA G SHETTY SRN : PES1UG24AM308 SEC : F

STEP 1: SCENARIO REVIEW

Scenario:

Digital Campus Library Reservation Gateway

Key Functional Requirements:

1. Search the ISBN catalogue and place hold requests on currently loaned books.

2. Maintain FIFO holding queues and notify the next member when a book becomes available.

3. Automatically calculate and update overdue fines.

4. Allow the Head Librarian to view, reorder, cancel, and override queue entries.

5. Allow students to pay outstanding fines online and restrict new holds when the balance exceeds ₹500.

Key Non-Functional Requirements:

1. Performance: Catalogue searches for 50,000 titles should return results in under 150 ms.

2. Security: Search and authentication endpoints must use HTTPS and role-based access control.

Main Challenges:

1. Performance – Fast catalogue searching and responsive queue operations.

2. Usability – Simple interfaces for students and librarians.

3. Security – Protect member data, authentication, and payment-related information.

STEP 2: ARCHITECTURAL STYLE SELECTION

Selected Architecture: Microservices Architecture

The Digital Campus Library Reservation Gateway will use a Microservices Architecture. The system can be divided into independent services such as catalogue search, reservation and queue management, notifications, fine management, and payment processing.

STEP 3: COMPONENT IDENTIFICATION

1. Catalogue & Reservation Service

   Handles ISBN catalogue search, book availability checking, and hold requests.

2. Order Manager

   Manages hold requests and coordinates reservation-related services.

3. Queue & Notification Service

   Manages FIFO holding queues, queue positions, and hold-availability notifications.

4. Fine Management Service

   Calculates overdue fines, maintains outstanding balances, and checks the ₹500 hold restriction.

5. Payment Service

   Processes online fine payments and returns payment status.

STEP 4: INTERFACES

1. Catalogue & Reservation Service → Order Manager

   Interface: Hold Request / Book Details

2. Order Manager → Queue & Notification Service

   Interface: Queue Management / Notification Request

3. Order Manager → Fine Management Service

   Interface: Fine Status / Eligibility Check

4. Order Manager → Payment Service

   Interface: Payment Processing Request

5. Payment Service → Fine Management Service

   Interface: Payment Status / Fine Update

STEP 5: ARCHITECTURE JUSTIFICATION

Selected Architectural Pattern: Microservices Architecture

The Digital Campus Library Reservation Gateway uses a Microservices Architecture because the system contains several independent functions such as library data management, reservation and queue management, notification services, and fine and payment services.The system is divided into separate components, allowing each service to perform a specific Component for searching books, while the Reservation & Queue Manager handles reservations and FIFO queues. The Notification Service informs users when books become available, and the Fine & Payment Service manages overdue fines and online payments.This architecture improves **scalability, maintainability, flexibility, and fault isolation**. For example, the catalogue search service can be scaled independently to support searches across 50,000 titles while maintaining the required response time. Similarly, changes to the payment or notification service can be made without affecting the entire system.The architecture also supports **security and role-based access control**, which is important because students and librarians have different permissions. Overall, Microservices Architecture is suitable because it separates responsibilities into independent services while allowing them to communicate through well-defined interfaces.

STEP 6: COMPONENT DIAGRAM

